

LESSON 5.8

TRIANGLES ON THE COORDINATE PLANE



LESSON INTRODUCTION

MA.8.GR.1.2 Apply the Pythagorean theorem to solve mathematical and real-world problems involving the distance between two points in a coordinate plane.

LEARNING TARGETS

- I can determine the horizontal and vertical lengths of a triangle on a coordinate plane.
- I can determine the hypotenuse length of a triangle on the coordinate plane.

BENCHMARK COHERENCE

BUILDING ON

MA.6.GR.1.2 Find distances between ordered pairs, limited to the same x -coordinate or the same y -coordinate, represented on the coordinate plane.

WORKING TOWARD

MA.9.12.GR.7.2 Given a mathematical or real-world context, derive and create the equation of a circle using key features.

ESSENTIAL QUESTIONS

- How can I use what I know about the Pythagorean theorem to help find perimeter of a triangle on the coordinate plane?

LESSON NARRATIVE

In this lesson, students begin to apply the Pythagorean theorem on the coordinate plane. Students will plot points that connect to create a right triangle. Then they will find the horizontal and vertical length visually before making the connection between distance and coordinate differences. Next, they will use these lengths and apply Pythagorean theorem to find the length of the hypotenuse. After this exploration, students will practice finding the lengths of triangles (including perimeter problems) using the Pythagorean theorem.

COMPONENT SUMMARY

5.8.1 WARM-UP (~5 MIN.)

Create true equations using all four operations

5.8.3 TRY IT (5 MIN.)

Find horizontal and vertical distance on coordinate plane

5.8.5 TRY IT (5 MIN.)

Find perimeter of right triangles on the coordinate plane

5.8.7 YOUR TURN! (5-10 MIN.)

Practice finding perimeter of triangles on the coordinate plane

5.8.2 GUIDED INSTRUCTION (10 MIN.)

Review finding distance/length on coordinate plane

5.8.4 GUIDED INSTRUCTION (10 MIN.)

Determine perimeter of right triangles on the coordinate plane

5.8.6 GUIDED PRACTICE (10 MIN.)

Find perimeter of triangles on the coordinate plane

5.8.8 WRAP-UP (~5 MIN.)

Self-reflection on lesson

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms:

- Isosceles
- Hypotenuse
- Pythagorean theorem

MATERIALS

- Calculators

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may make errors when finding the vertical or horizontal distances on the coordinate plane.
- Students may plot points incorrectly, mixing up the x- and y-values.

THINGS TO CONSIDER

- Consider having students write out the formula for the Pythagorean theorem before answering each question. This should help students visualize substituting each number.
- Consider having students color-code the sides of the triangle a, b and c, each corresponding to a certain color.
- If time is an issue, then consider assigning 5.8.7 Your Turn! for homework.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

In 5.8.2 Guided Instruction, students make observations about the vertical and horizontal distances. Prompt students to engage in a Notice and Wonder about these distances. This will help students draw conclusions about finding length on the coordinate plane.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.6.1 Reasonableness

Students have opportunities throughout this lesson to check the reasonableness of their solutions. These problems require multiple steps, which could result in a mistake along the way. Students can check for mistakes by checking the reasonableness of their solutions using the theorems learned previously in this unit.

DIFFERENTIATE INSTRUCTION

Consider using the On-Ramp to 6th Grade learning tool to provide scaffolding for students who may need additional support plotting points on the coordinate plane. It may be helpful to utilize these videos as a preview to the lesson or have them available on a computer for students to reference for extra help.

5.8.1 WARM-UP: CRITICAL THINKING

TEACHER MOVES

Note there are multiple solutions for each answer. Students may forget to use the order of operations. Consider a reminder if needed.



5.8.1 Warm-Up: Critical Thinking

1. Use any combination of operations (+, −, ×, ÷) to arrange five 5's to have an equivalent value of 0, 1, 2, 3, 4, and 5. Multiple 5's can be used in one number (e.g. 55).

Example: $(5 \times 5 - 5 \times 5) \times 5 = 0$

Answers will vary.

$$0 = 5(5 - 5 - 5 + 5) \qquad 3 = (5 - (5 \div 5)) - (5 \div 5)$$

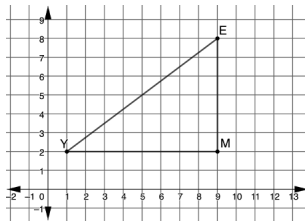
$$1 = (5 \div 5) - (5 \cdot (5 - 5)) \qquad 4 = (5 + 5 - 5) - (5 \div 5)$$

$$2 = (55 - 5) \div (5 \cdot 5) \qquad 5 = 5 + 5 - 5 - 5 + 5$$



5.8.2 Guided Instruction: Finding the Length

The Triangle EMY is shown on the coordinate plane.



- List the ordered pairs for each vertex.
 - Point E (9 , 8)
 - Point M (9 , 2)
 - Point Y (1 , 2)
- What is the length of the horizontal leg of Triangle EMY ?
8 units
- What do you notice about the horizontal length and the coordinates of point M and point Y ?
Answers will vary. I notice that the horizontal length is equal to the difference between the x-coordinates of point M and point Y .
- What is length of the vertical leg of Triangle EMY ?
6 units

- What do you notice about the vertical length and the coordinates of points E and points M ?
Answers will vary. I notice that the vertical length is equal to the difference between the y-coordinates of point E and point M .

- Complete the statements.

To determine horizontal length on the coordinate plane,

find the difference between the

- ☒ x-coordinates.
- ☐ y-coordinates.

To determine vertical length on the coordinate plane,

find the difference between the

- ☐ x-coordinates.
- ☒ y-coordinates.

5.8.2 GUIDED INSTRUCTION: FINDING THE LENGTH

TEACHER MOVES

This lesson is scaffolded to allow students to review concepts learned in 6th grade, then apply the Pythagorean theorem to find the length of the hypotenuse of a triangle on the coordinate plane. Please note that although students are finding the distance between two points, there is no mention of a distance formula. Students often find the distance formula complicated, while applying the Pythagorean theorem is often easier for students. The distance formula compiles all of the steps in this lesson into a formula. So, if students understand the steps presented here, they will not need the formula.

DIFFERENTIATE INSTRUCTION

Consider projecting Triangle EMY for the class to see. Providing an additional larger visual of the image reduces visual barriers to entry through flexible displays of the information.

TEACHER MOVES

Common Misconception

If students say the distance is 9, they may be counting the coordinate lines instead of the distance between them.

5.8.3 TRY IT: FINDING HORIZONTAL AND VERTICAL DISTANCES

DIFFERENTIATE INSTRUCTION

Consider having students highlight the x-coordinates in one color and the y-coordinates in another. Students will benefit from seeing which coordinate's values (x or y) are different. Those are the values that should be subtracted to find the length of that side.



5.8.3 Try It: Finding Horizontal and Vertical Distances

1. Find the **horizontal** distance between the following points.
 - a. $(2, 7)$ and $(8, 7)$ 6 unit(s)
 - b. $(-5, 1)$ and $(3, 1)$ 8 unit(s)
 - c. $(0, 4)$ and $(6, -3)$ 6 unit(s)

2. Find the **vertical** distance between the following points.
 - a. $(2, -9)$ and $(2, 6)$ 15 unit(s)
 - b. $(-4, -1)$ and $(-4, -3)$ 2 unit(s)
 - c. $(0, 4)$ and $(6, -3)$ 7 unit(s)

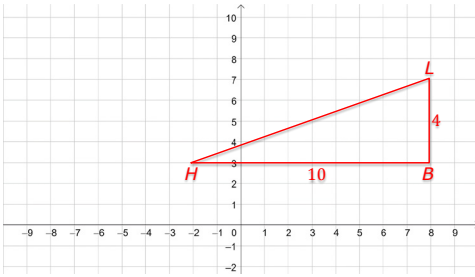


5.8.4 Guided Instruction: Right Triangles on the Coordinate Plane

1. A triangle is defined by the following vertices in the coordinate plane.

Point $B (8, 3)$ Point $H (-2, 3)$ Point $L (8, 7)$

- a. Graph Triangle BHL .



- b. Find the length of the horizontal leg and label the leg with its length.
10 units
- c. Find the length of the vertical leg and label the leg with its length.
4 units
- d. Using the Pythagorean Theorem, find the length of the hypotenuse (HL). Round to the nearest tenth, if necessary.
10.8 units
- e. Find the perimeter of the triangle.
24.8 units

5.8.4 GUIDED INSTRUCTION: RIGHT TRIANGLES ON THE COORDINATE PLANE

TEACHER MOVES

Consider having students label the sides of the triangle with a , b , and c . If students are struggling to plot the points or find the lengths of the legs, prompt with questions.

- What information are we given? How can you show this on the graph?
- How can you find the length of the horizontal leg? Which coordinates should you find the difference between?
- How can you find the length of the vertical leg?

DIFFERENTIATE INSTRUCTION

Consider having the Pythagorean theorem posted somewhere in your room for students to reference. Encourage students to write out the theorem as they find the length of the hypotenuse to reduce error and encourage precision.

5.8.5 TRY IT: RIGHT TRIANGLES ON THE COORDINATE PLANE

ENRICHMENT

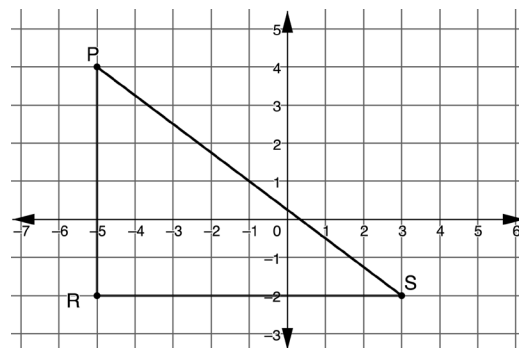
Some students may try to count the spaces of the hypotenuse and may therefore find the incorrect length. If students struggle to find the perimeter, further scaffold this section through questioning.

- What information do we need to find the perimeter?
- Do the horizontal and vertical lengths represent the legs or hypotenuse? How do you know?
- How can you find the length of the legs?
- How can you use the length of the legs to find the length of the hypotenuse?



5.8.5 Try It: Right Triangles on the Coordinate Plane

Triangle PRS is graphed on the coordinate plane.



1. Find the perimeter of Triangle PRS . Round to the nearest tenth, if necessary.

24 units



5.8.6 Guided Practice: Finding Perimeter

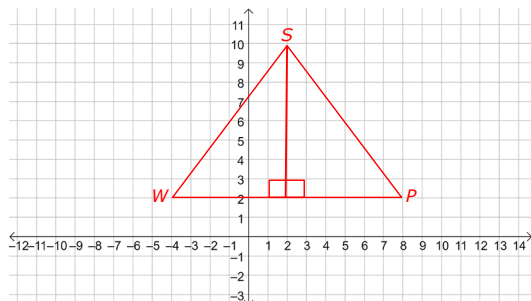
1. A triangle is defined by the following vertices on the coordinate plane.

Point W $(-4, 2)$

Point P $(8, 2)$

Point S $(2, 10)$

- a. Graph Triangle WPS .



- b. What type of triangle is Triangle WPS ?
Isosceles
- c. Find the horizontal length of Triangle WPS .
12 units
- d. Draw a perpendicular line segment from point S to WP .
- e. Use the Pythagorean Theorem to find the lengths of WS and PS .
10 units; 10 units
- f. Find the perimeter of Triangle WPS .
32 units

5.8.6 GUIDED PRACTICE: FINDING PERIMETER

DIFFERENTIATE INSTRUCTION

Consider a review of the vocabulary for the different types of triangles and what they look like to activate prior knowledge and provide additional entry points for students.

TEACHER MOVES

Students may try to use 12 as one of the legs of the right triangles. After drawing a perpendicular line, consider having students write the horizontal distance of the two right triangles that have been created to prevent students from using 12.

ENRICHMENT

Students may not realize they don't have all the information they need to use the Pythagorean theorem. Prompt students who make this mistake.

- Do you have all the information you need to use the theorem? How can you label the triangle to determine this?
- What other piece of information do we need to find first before we can use the Pythagorean theorem? (The vertical length.)

5.8.7 YOUR TURN!

DIFFERENTIATE INSTRUCTION

To provide additional entry points and another means of engagement, consider having students plot the points using the Desmos graphing calculator available online.

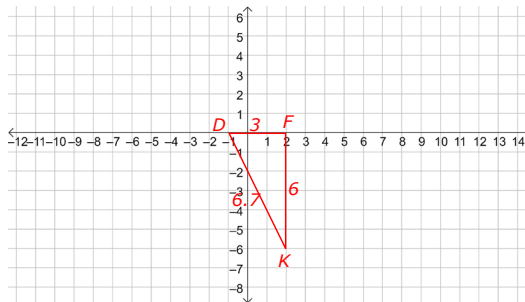


5.8.7 Your Turn!

1. Triangle DKF is defined by the following vertices.

$$D(-1,0) \quad K(2,-6) \quad F(2,0)$$

- a. Graph Triangle DKF .



- b. Label each side with its length. Round to the nearest tenth, if needed.

TEACHER MOVES

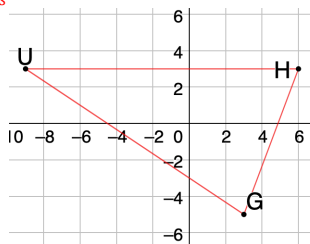
Encourage those students that may be stuck to graph the triangle first and label the side lengths so they can easily be tallied later.

2. Triangle UHG is defined by the vertices.

$$U(-9,3) \quad H(6,3) \quad G(3,-5)$$

Find the perimeter of Triangle UHG . Round to the nearest tenth, if needed.

38 units





5.8.8 Wrap-Up: Reflection

1. Place an **X** on each line to indicate your understanding of each success criteria.
Answers will vary.

Success Criteria	Self-Assessment
I can apply the Pythagorean Theorem to find the distance of a segment on the coordinate plane.	I need help. I can't get started. I'm getting there. I understand and I have evidence.
I can find the perimeter of a right triangle on the coordinate plane.	I need help. I can't get started. I'm getting there. I understand and I have evidence.

5.8.6 GUIDED PRACTICE: FINDING PERIMETER

QUESTIONING

For students who finish early, consider having them indicate points throughout the lesson to illustrate their self-assessments of each statement.